

Ariel Morillo

Quito, Ecuador

arielmorillo.ct@gmail.com | +593 962 526 475

linkedin.com/in/ariel-morillo-dev | github.com/arieleMM-dev

Motivation Letter for an International Software Engineering Internship

Dear Selection Committee,

I am a Software Engineering student at Universidad de las Fuerzas Armadas ESPE, building upon a foundational background in Telecommunications Engineering. I am seeking a software engineering internship at CERN to contribute to the resilient, large-scale systems that empower global scientific research.

My technical focus lies in backend engineering and system architecture. Recently, I led the backend development for Finca Jirah, an agricultural traceability platform. Beyond building REST APIs with Next.js and modeling relational data via Prisma ORM and PostgreSQL, I engineered an Offline-First architecture using IndexedDB. Implementing deferred synchronization for areas with limited connectivity taught me how to architect for network failure and data consistency under unpredictable real-world conditions.

Furthermore, my experience designing ETL pipelines with Pentaho and SQL Server has solidified my understanding of data lifecycle management, while my background in Telecommunications provides me with a deep appreciation for the underlying infrastructure and network layers that distributed software relies on.

I am driven by the challenge of building fault-tolerant backend services. An internship at CERN represents an unparalleled opportunity to learn how world-class research teams develop, validate, and maintain services that process massive data workloads. I bring a strong foundation in JavaScript/TypeScript ecosystems, a commitment to clean architecture, and a readiness to absorb feedback and tackle complex engineering problems.

Thank you for considering my application. I look forward to the possibility of discussing how my technical background aligns with your team's objectives.

Yours faithfully,

Ariel Morillo